Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/15/2020
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Project Title:	Synergistic Study of Radioprotection Mechanisms Between Flies and Human Hippocampal Neurons		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Comments:	Formerly at Emory University, relocated in September 2014 (Ed., 7/7/15)		
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Synergy
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fox, Donald Ph.D. (Duke University Medical Center)		
Grant/Contract No.:	NNX16AO69A-SYN0002		
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Synergy Project
In the Translational Research Institute for Space Health (TRISH)-funded study titled "Mining biological extremes for new space radiation resistance strategies" by Dr. Fox, peer reviewers raised two challenges: 1) Single-gene dominant effects that transfer radio-resistance are presumed, 2) If a protein is to be expressed in astronauts, it would be nice to know if it affects cognition or is toxic in some other way. Here, we propose to address these challenges, by: 1) Developing Drosophila as an in vivo model to study known radioprotectors found in cultured human neurons, and test whether these human cell radioprotectors synergize with radioprotector genes found in Drosophila (human cells to flies); 2) Assessing the effects of newly identified extreme radioresistant genes using a hippocampal neuron model (flies to human cells).

We had revealed several potential radiation protectors and neural protectors including CDK5 inhibitors, GSK3 inhibitor,

Task Description:	and PARP inhibitors. We will use Drosophila to assess organism-wide and central nervous system (CNS)-specific effects of these inhibitors in vivo. Sept 2018-Mar 2019: Upon identification of new radioprotectors found in Drosophila, we will examine the role of such genes in human cultured hippocampal neuron model, in response to low and high linear energy transfer (LET) radiation, and will also use Drosophila to assess whether the above-mentioned inhibitors synergize with newly identified radioprotector genes. We will perform the following experiments: Experiment 1. To detect radiation protection of CDK5 inhibitor, GSK3b inhibitors, and PARP inhibitor on radiation sensitivity and life span in Drosophila. This experiment will be cooperated with experiments in Aim1b of Dr. Fox's grant to test radiation resistance in adult flies. First we will determine the optimal dose range of X-ray and HZE (high energy) particles in flies by testing dose response of fly viability. We will use 100 animals per dose (50 male and to 50 female). Numbers of surviving irradiated flies will be scored at one week, two week, and three weeks after irradiation. The survival rate is relative of irradiated number to unirrradiated number. Then we will test different concentration of inhibitors in one dose. The flies will be treated with inhibitors before irradiation. The relative biological effects and the protection ratio will be assessed. Second, we will examine the fecundity of the flies. 10 female flies will be mated immediately after irradiation. Over a period of 3 weeks post irradiation, the number of viable adult F1 progeny of individual irradiated flies will be scored. To monitor multi-generational fecundity, a similar number of individual number of individual F1 adults will be similarly monitored for viable F2 Progeny. Third, we will examine DNA damage and genomic instability in irradiated flies. The irradiated flies will be fixed and stained with ant-gamma H2AX, anti-phosphoryated ATM s1981, and anti-phosphorlated CREB. DNA damage foci will be observed under confocal microscopy Leica Sp8 and analyzed by imaging software Imaris. The irradiated flies will be Dr. Fox's work and will be performed in Duke University. Experiment 2. To examine the role of new identified genes from Drosophila in human neurons using Dr. Wang's cultured hippocampal mode. We will use an immortalized human neuroblast cells and differentiated neurons from human iPSC (induced pluripotent stem cell) cells to detect of the effect of new genes in radiation sensitivity, DNA damage, and neural functions. We will detect the interaction of new gene expression with human DNA repair proteins (including Ku, RPA, Rad51, Mre11, PARP, and ATM) and their functions in human hippocampal neurons. This work will be performed in Prairie View A&M University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In this study, we developed a novel model of human neurons and cardiomyocytes derived human induced pluripotent stem cell expressing a sensitive calcium indicator to discover radioresistant factors and develop radiopotectors for space health.
Task Progress:	[NOTE (Ed., 5/15/2020): Report submitted to TRISH in April 2019] The major risk from space environment includes microgravity and space radiation (Proton and high energy charge particles (HZE) from solar particle event and galactic cosmic ray (GCR)). Evidence from space radiobiological studies points out that HZE and protons induced the central nervous behavior change leading to behavior change, aging process, neurodegenerative disease, and neurogenesis defects caused from persistent DNA damage, oxidative stress, and inflammation, based on the animal models. The big challenge is how to incorporate these stressors and translate useful data from animal models to predict the risk to human performance. In the TRISH-funded study titled "Mining biological extremes for new space radiation resistance strategies" by Dr. Fox, peer reviewers raised two challenges: 1) Single-gene dominant effects that transfer radio-resistance are presumed, 2) If a protein is to be expressed in astronauts, it would be nice to know if it affects cognition or is toxic in some other way. Here, we propose to address these challenges by: 1) Developing Drosophila as an in vivo model to study known radioprotectors found in cultured human neurons, and test whether these human cell radioprotectors synergize with radioprotector genes found in Drosophila (human cells to flies); 2) Assessing the effects of newly identified extreme radioresistant genes using a hippocampal neuron model (flies to human cells). We established a cultured human neuron system derived from human induced human induced pluripotent stem cell (HiPSC) for discovering radioresistant factors and developed radioprotectors for space health. We genetically transduced calcium indicator (GcAMP6f) in HiPSC derived human neurons for neuronal function. We found that Nicotinamide ribose (NR) and nicotinamide mononucleitide (NMN) protect human neurons from radiation. We examined the effect of inhibitors of CDK5, GSK3beta, and PARP on radiation sensitivity and life span in Drosophila.
Bibliography Type:	Description: (Last Updated: 11/13/2019)